

# HARISH KATHIRAVAN M

AI Engineer · Machine Learning · Computer Vision · Full Stack Development

Tamil Nadu, India · [github.com/harishkathiravan-sys](https://github.com/harishkathiravan-sys) · [www.linkedin.com/in/harish-kathiravan-05b229304](https://www.linkedin.com/in/harish-kathiravan-05b229304)

## PROFESSIONAL SUMMARY

B.Tech CSE student with hands-on experience building end-to-end AI applications spanning EEG emotion recognition, real-time computer vision, NLP-based threat detection, and full stack web/mobile development. Delivered six production-grade projects integrating ML models with real-world interfaces using Python, React, and Flutter. Focused on generative AI, explainable ML, and scalable model deployment.

## TECHNICAL SKILLS

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Languages         | Python, JavaScript, TypeScript, Java, SQL, Dart                                                          |
| AI / ML           | Machine Learning, Deep Learning, Computer Vision, NLP, Explainable AI, Prompt Engineering, Generative AI |
| Frameworks        | Scikit-learn, OpenCV, MediaPipe, LangChain, TensorFlow, React, Next.js, Node.js, Express.js, Flutter     |
| Databases & Tools | MongoDB, MySQL · Git, GitHub, VS Code, Postman, Vercel, Render                                           |

## FEATURED PROJECTS

### EEG Bot · *Emotion-aware AI assistant via EEG signal analysis*

- ▶ Built an EEG signal pipeline classifying five emotional states with XGBoost; integrated SHAP and LIME for transparent, clinically interpretable model decisions.
- ▶ Developed a LangChain conversational backend adapting responses to detected emotional context, delivered via a React + Node.js interface.  
**Tech:** Python, XGBoost, SHAP, LIME, LangChain, React, Node.js

### RajiniVision AI · *Real-time hand gesture recognition*

- ▶ Engineered a live video gesture classification pipeline with OpenCV and MediaPipe, achieving 30+ FPS inference for custom gesture vocabularies.  
**Tech:** Python, OpenCV, MediaPipe, Computer Vision, Machine Learning

### MailShield AI · *AI-based phishing and email threat detection*

- ▶ Applied TF-IDF and ensemble ML models to classify phishing, spam, and legitimate emails; implemented risk-scoring with actionable user-facing explanations.  
**Tech:** Python, NLP, Scikit-learn, Machine Learning, REST APIs

### Complaint Routing System · *NLP-driven complaint classification platform*

- ▶ Built an NLP text classifier that auto-routes complaints to the correct department; delivered a React admin dashboard with real-time status tracking via REST APIs.  
**Tech:** Python, NLP, Machine Learning, React, Node.js, MongoDB

### TravelBuddy · *AI-assisted cross-platform travel companion*

- ▶ Developed a Flutter mobile app with AI-driven itinerary recommendations and natural language trip planning via integrated conversational AI features.  
**Tech:** Flutter, Dart, AI Integration, REST APIs

### KwikKart · *Full stack e-commerce platform*

- ▶ Built end-to-end e-commerce app with JWT auth, product CRUD, and order management; deployed via Vercel and Render with GitHub-based CI/CD.  
**Tech:** React, Node.js, Express.js, MongoDB, JWT, Vercel, Render

## WORK EXPERIENCE

### Web Development Intern · GreenWill Techs, Tamil Nadu

- ▶ Built responsive React interfaces and RESTful Node.js/Express.js APIs, delivering production-ready features for real clients.
- ▶ Collaborated using Git and GitHub in an agile team, practicing code reviews and iterative deployment workflows.

## EDUCATION

### Bachelor of Technology — Computer Science Engineering · Tamil Nadu, India

## ACHIEVEMENTS & GITHUB

- ▶ Completed industry internship at GreenWill Techs, contributing to production web applications.
- ▶ Developed and shipped six end-to-end AI and full stack projects covering EEG processing, computer vision, NLP, and mobile development.
- ▶ Active GitHub profile with repositories across AI, Computer Vision, NLP, Flutter, and Full Stack Development — demonstrating consistent, practical engineering.
- ▶ Independent research in Explainable AI: applied SHAP and LIME to interpret ML model decisions for domain experts.